

48 | OVERVIEW



Tracking Record Ocean Heat and Extreme Weather

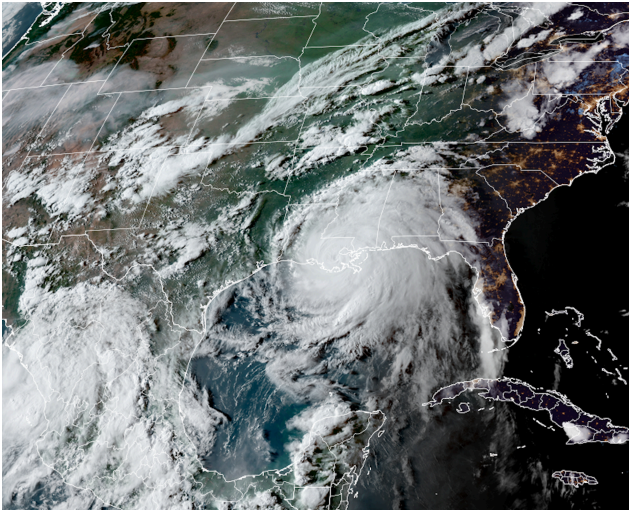
Abstract

In 2021, the world's oceans were hotter than ever recorded, even though surface air temperatures ranked only fifth or sixth warmest. In an interview, climate scientist Kevin Trenberth, a coauthor of a 2022 study by 23 researchers at 14 institutes, explained why ocean heat is a better measure of global warming and how it supercharged storms, floods, heat waves, and wildfires.

Introduction

2021 brought record-breaking climate and weather disasters: flash floods in European mountain towns, flooded subway systems in China and the U.S., heat waves, and wildfires. Typhoon Rai killed over 400 people in the Philippines, and Hurricane Ida caused an estimated \$74 billion in damage in the U.S. Yet NASA and NOAA ranked 2021 only the sixth hottest year for surface temperatures, partly because of year-long La Niña conditions. Beneath the surface, however, the oceans set new heat records. Trenberth explains that while we feel surface temperature day to day, the upper ocean is the planet's primary memory of global warming.

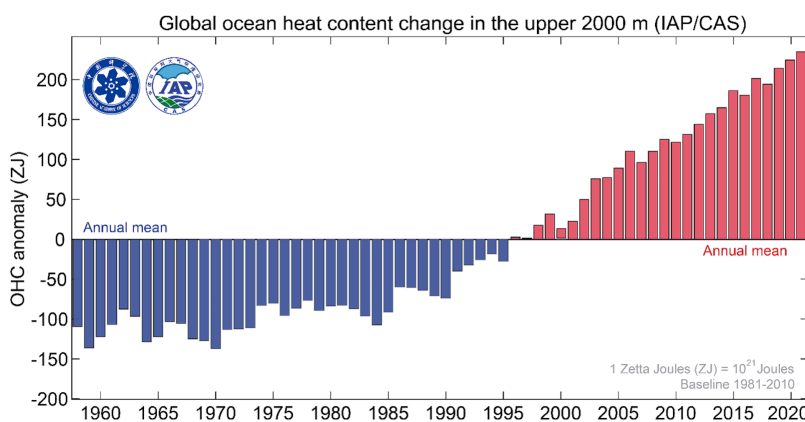
How Do Scientists Study This?



FROM THE ARTICLE

Hurricane Ida did \$74 billion in damage from Louisiana to the northeastern U.S. in 2021. Image: RAMMB/CIRA/Colorado State University.

The study tracked ocean heat content in the upper 2,000 meters of the world's oceans from 1958 to 2021, comparing each year with the 1981-2010 average. Much of the recent data comes from the Argo array: about 3,900 robotic floats that drift through ocean basins, diving and rising to measure temperature and salinity from the surface down to about 2,000 meters. Researchers analyzed warming by ocean basin and by depth layer to see where heat is going. They also used their analysis to attribute the warming to human activities, since the oceans absorb about 93% of the extra energy trapped by greenhouse gases.



FROM THE ARTICLE

Ocean heat content in the upper 2,000 meters since 1958, relative to 1981-2010, in zettajoules. Image: Lijing Cheng.

Explore the Evidence

How ocean heat drives extremes

1

Greenhouse gases

Trap extra energy

2

Oceans absorb

About 93% of the extra heat

3

More moisture

Evaporation fuels storms

4

Extremes

Hurricanes, floods, marine heat waves

Deploy floats

About 3,900 Argo floats measure temperature and salinity down to 2,000 meters.

Measure by layer

Scientists compare warming at different depths and in different ocean basins.

Attribute the cause

The analysis linked record ocean warming to human activities.

What Did They Discover?

Ocean heat has increased every decade since the 1960s, a primary indicator of human-caused climate change. All oceans are warming, most of all the Atlantic and the Southern Ocean around Antarctica, where heat can creep under ice shelves, thinning them and causing huge icebergs to break off. The top 500 meters has warmed since 1980, 500 to 1,000 meters since about 1990, 1,000 to 1,500 meters since 1998, and below 1,500 meters since about 2005. Warmer oceans supply extra moisture that fuels hurricanes and atmospheric rivers, causing prodigious rainfall and floods. Because the ocean warms from the top down, it is becoming more stratified, reducing mixing of heat, carbon dioxide, and oxygen and affecting all marine life.

Ocean heat in 2021

93%

of trapped energy taken up by the oceans

3,900

Argo floats measuring down to 2,000 meters

\$74B

damage from Hurricane Ida in the U.S.

Because heat penetrates the ocean slowly, oceans will keep warming and seas will keep rising even after greenhouse gases are stabilized.



FROM THE ARTICLE

Scientists are concerned about the stability of Antarctica's Thwaites Glacier. Image: NASA.

48 | TAKEAWAYS

Takeaways

Oceans hold the record

Ocean heat broke records in 2021 even when surface temperatures did not.

Heat fuels extremes

Warm oceans supercharge storms, rainfall, and flooding.

Monitor the depths

Robotic Argo floats are essential for tracking ocean change.

Key Terms

Ocean heat content

The total heat stored in the ocean, measured in zettajoules, a primary indicator of global warming.

Argo float

A robotic instrument that drifts, dives, and rises through the ocean to measure temperature and salinity.

Marine heat wave

A period of unusually warm ocean temperatures in a region, harmful to marine life.

Stratification

Layering of ocean water by temperature and density, which limits mixing between layers.

Atmospheric river

A long, narrow band of moisture in the atmosphere that can bring heavy rain or snow when it reaches land.

48 | ACTIVITY

Activity

Layers of the Warming Ocean

Students build a timeline showing how warming has reached deeper ocean layers over time and model stratification.

Materials: Tall clear container; Warm and cold water with food coloring; Timeline strip

1. Create a vertical timeline showing when each depth began warming: 0-500 m (1980), 500-1,000 m (~1990), 1,000-1,500 m (1998), below 1,500 m (~2005).
2. Gently pour colored warm water on top of cold water in the container. Observe whether they mix.
3. Explain how this models ocean stratification and why it limits oxygen reaching deeper water.
4. Discuss why the oceans will keep warming after emissions stabilize.
5. Write one reason ocean heat is a better indicator of global warming than one year's air temperature.

References

Trenberth, K. (2022). Ocean heat is at record levels, with major consequences. Interview in The Conversation. <https://doi.org/10.64628/AAL.fyf4c733t>

Ocean warming study published January 11, 2022, by 23 researchers at 14 institutes, coauthored by Kevin Trenberth, as described in the article above.